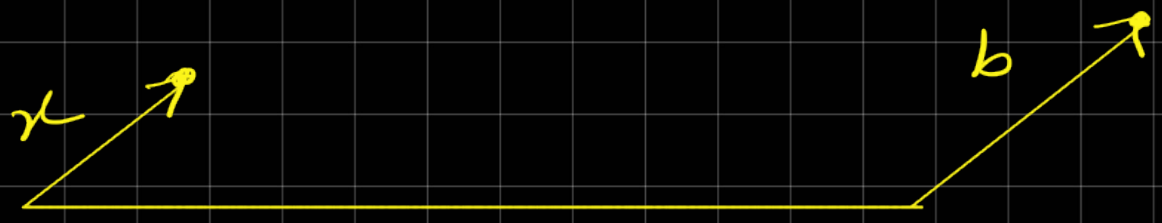
Eigenvectors & Eigenvalues

A mat takes 1 vector x spits out another vector b.



$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \xrightarrow{\text{scales by 2}}$$



Ax $\xrightarrow{\text{resulting vectors}}$ is in parallel to x (same direction)

↳ Then x is Eigenvector.

$Ax = \lambda x$ → direction same

(λ: Eigenvalue)

↳ Eigenvector

scaling factor

if $\lambda = 0$ then $Ax = 0$

eqn of Nullspace

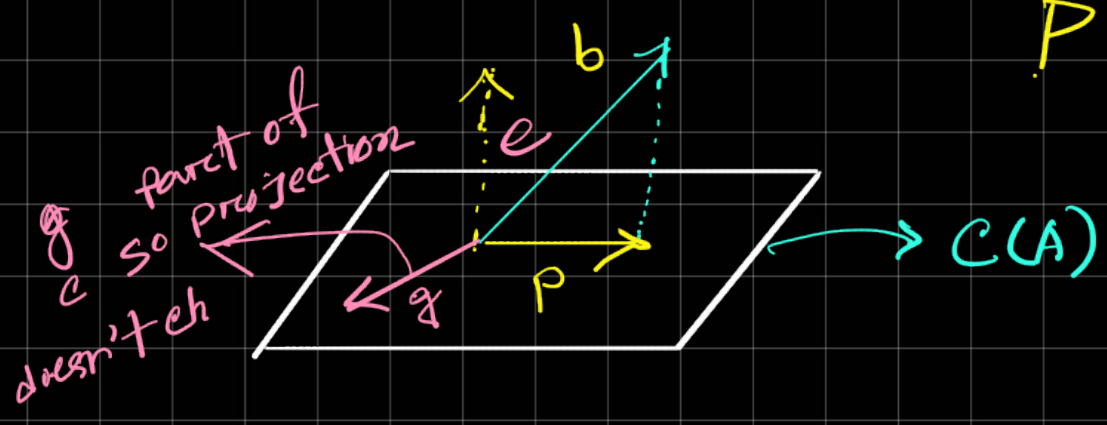
x belongs to Nullspace of A.

$Ax = 0$

$\rightarrow x \in N(A)$

So scaled by 0. Any vector in Nullspace is Eigenvector with Eigenvalue 0.

Every Mat has 0 vector as E.V.



b can't be E.V, not part of plane + changes direction.

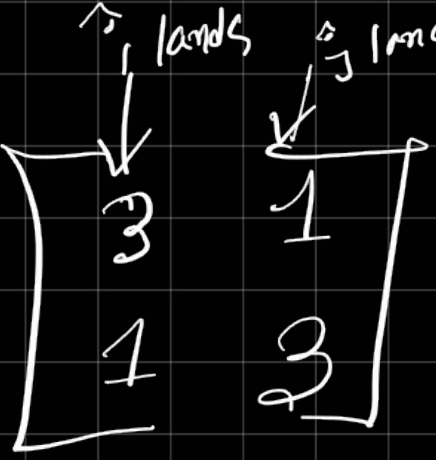
p doesn't chang direction so

Eiv of A. scaled by 1.

All vectors in C(A) is Eigen Vec of A of Eigenvalue of 1.

Eigen vector.
Hiding in plain sight e.. where $\lambda = 0$.

So the vectors perpendicular to C(A) is eigenvector of A, meaning every vector in left Nullspace is A's E.V. with the Eival 0.



2x2

→ how many Eigen Vectors can be found
→ Actually infinite but independent
→ (2) usually

$(-1, 2) \rightarrow (-2, 4)$

indep. dep

$$\begin{bmatrix} 3 & 1 & 3 \\ 1 & 3 & -2 \end{bmatrix} \begin{bmatrix} \quad \\ \quad \\ \quad \end{bmatrix} = \begin{bmatrix} \quad \\ \quad \end{bmatrix}$$

2×3 $\quad$ 3×1 $\quad$ 2×1

takes 3 dimensional vectors spits out 2D vectors.

So dimension changed.

Impossible to compare 2D & 3D vectors & find whether they point to the same direction or not.

So, Non sq Mat don't have eigenvectors.

$$Ax = \lambda x$$

$$\Rightarrow (Ax - \lambda x) = 0 \qquad Ax = 0$$

$$\Rightarrow \underbrace{(A - \lambda I)}_{B}\, x = 0$$

can't write $(A-\lambda)$ cz λ scalar so Mat − scalar no possible

Solve So we can get nullspace.

$(A-\lambda I)\,x$ → 2 unknown, 1 eqn → can't solve

B

↓

So be can't be invertible

there must be no zero pivots.

if invertible then det = 0. (not interested in 0 vects

$|A - \lambda I| = 0$

already EV of old matrix

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow (3-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 6\lambda + 9 - 1 = 0$$

$$\boxed{\lambda^2 - 6\lambda + 8 = 0}$$

$A - \lambda I$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2b}$$

will give me problems.

$$(\lambda - 4)(\lambda - 2) = 0$$

$\lambda_1 = 4 \quad | \quad \lambda_2 = 2$

$\lambda = 4$ | $\lambda = 2$

$$\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} x=0 \qquad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x=0$$

1 times 1st col
1 " 2nd col

$$x=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$$x=\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Random
If any^ Matrix is given, will it have E.vec & E.val?

↳ Ans: Most likely complex value. ✓

complex ↗
|| Eigen_vect.

If I rotate, Eval & E vec work together. all vectors change direction.

$$A=\begin{bmatrix} 0 & -1 \end{bmatrix}$$

$\begin{bmatrix} 1 & 0 \end{bmatrix}$

↳ if symmetric Mat, always real E.val, E.vec.

(90° always) [are ⊥ to each other] ←

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\begin{vmatrix} 3-\lambda & 1 \\ 0 & 3-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)^2 = 0$$

$\lambda_1 = 3 \quad \lambda_2 = 3$

prblm

takim $\lambda = 3$

$$\begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} x = 0$$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $= \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

If A^{-1} exists, what is E.val?

$Ax = \lambda x$

$A^{-1}Ax = A^{-1}\lambda x$

$\Rightarrow x = \lambda A^{-1}x$

$\Rightarrow A^{-1}x = \frac{1}{\lambda}x$

→ λ can't be zero.

So, when can't λ be zero?